

Effect of angina under-recognition on treatment in outpatients with stable ischaemic heart disease

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Aims

Almost a third of outpatients with chronic coronary artery disease (CAD) report having angina in the prior month, which is frequently under-recognized by their cardiologists. Whether under-recognition is associated with less treatment escalation to control angina, and potential underuse of treatment, is unknown.

Methods and results

Patients with CAD from 25 US cardiology outpatient practices completed the Seattle Angina Questionnaire (SAQ) prior to their clinic visit, and angina was categorized as daily, weekly, monthly, and no angina. Cardiologists ($n = 155$) independently quantified patients' angina, blinded to patients' SAQ scores. Under-recognition was defined as the physician reporting a lower category of angina frequency than the patient. Among 1257 patients with CAD, 411 reported angina in the past month, of whom 178 (43.3%) patients were under-recognized. Treatment escalation—defined as intensification (up-titration or addition) of antianginal medications, referral for diagnostic testing or revascularization, or hospital admission—occurred in 106 (25.8%) patients with angina. Patients with under-recognized angina were less likely to get treatment escalation than patients whose angina was appropriately recognized (8.4 vs. 39.1%, $P < 0.001$). In a hierarchical multivariable logistic regression model adjusting for demographic and clinical characteristics, as well as the burden of angina, under-recognition remained strongly associated with a lack of treatment escalation (adjusted OR 0.10, 95% CI 0.04–0.21, $P < 0.001$).

Conclusion

Under-recognition of angina in cardiology outpatient practices is associated with less aggressive treatment escalation and may lead to poorer angina control. Standardizing clinical recognition of angina using validated tools could reduce under-recognition of angina, facilitate treatment, and potentially improve outcomes.

Keywords

Coronary artery disease • Angina • Antianginal medications • Quality of care

Introduction

Among patients with coronary artery disease (CAD), angina is not only common¹ but also can adversely affect patient's quality of life.^{2,3} Consequently, eliminating angina and improving quality of life are primary treatment goals for the outpatient management of stable CAD.⁴ Treatment of angina can be achieved through multiple modalities including medications^{4–8} and revascularization.^{3,9–13} Even among patients with continued angina despite maximal medical therapy and whose anatomy is not amenable to revascularization, options for treatment exist in the form of spinal cord stimulation,¹⁴

enhanced external counter pulsation,^{15,16} and other novel strategies.^{14,17–19} However, in order to select the appropriate treatment plan for the patient, the physician must first appropriately recognize the patient's angina burden.

Due to the inherent limitations of history taking, and the time constraints of patients and physicians, appropriate recognition of angina can be challenging.²⁰ In a recent study of patients with CAD and angina, we found that 42% of patients reported more frequent angina than the cardiologists estimated that they had experienced, with substantial variability in the under-recognition rate across providers (0–85%).²¹ While this high rate of under-recognition of

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angina among outpatients with CAD is concerning, the clinical implications are unknown.

We hypothesized that if physicians under-recognized the frequency of patients' angina, they would be less likely to intensify antianginal treatment. As such, we examined factors that are associated with treatment escalation (either intensification of antianginal medications or referral for diagnostic or invasive testing) and, specifically, whether under-recognition of angina was associated with a lack of treatment escalation. Finding evidence of a lack of treatment escalation among patients with under-recognized angina could suggest missed opportunities to improve patients' health status and under-use of potentially beneficially therapies. It would also support the use of standardized, patient-reported assessments for angina among outpatients with CAD as a potential strategy to minimize under-recognition in routine clinical practice.

Methods

Study design and population

The Angina Prevalence and Provider Evaluation of Angina Relief (APPEAR) study was a cross-sectional, observational study designed to assess angina and its effect on quality of life in outpatients with CAD. The 25 participating sites were identified from US cardiology outpatient practices participating in the PINNACLE Registry (a national registry sponsored by the American College of Cardiology). For 1–2 weeks of active enrolment per site, between April 2013 and July 2015, local study coordinators recruited 50–75 consecutive adults with a history of CAD; defined as a diagnosis of stable angina, prior myocardial infarction, prior percutaneous coronary intervention (PCI), or prior coronary artery bypass graft surgery. Sites were instructed to enrol consecutive patients with CAD, regardless of the reason for their clinic visit. Eligible patients included those ≥ 18 years of age who had at least one prior office visit to the practice. Patients with dementia, those unable to speak or read English, and prisoners were excluded. Each participating site obtained Institutional Research Board approval, which allowed for a waiver of written informed consent at all sites, except one where patients provided written informed consent.

Assessment of angina and definition of under-recognition

A unique feature of angina is that laboratory and imaging tests cannot measure it. As such, there is no 'gold standard' to capture patients' experiences.²² Patient-reported symptoms of angina are strongly associated with patients' quality of life,²³ are predictive of hospitalizations for acute coronary syndromes and mortality,²⁴ and are strong drivers of healthcare utilization.²⁵ Thus, we believe angina is a symptom experienced by the patient and best reported by the patient. Accordingly, enrolled patients were asked immediately prior to their cardiologist visit to complete the Seattle Angina Questionnaire (SAQ). The SAQ is a self-administered, reliable, and valid 19-item questionnaire with a 4-week recall period that measures five domains of health in patients with CAD: Angina Frequency (SAQ AF), Angina Stability, Quality of Life, Physical Limitation, and Treatment Satisfaction.^{26,27} Additionally, the SAQ Summary Scale combines the three domains of symptoms, function, and quality of life, in order to integrate the multiple SAQ domains into a single score.²⁸ Individual domain scores and the Summary Score range from 0 to 100, with higher scores indicating fewer symptoms and better quality of life. The SAQ has undergone extensive reliability and validity testing^{27,28} and is associated with long-term survival, hospitalization for

acute coronary syndromes, and increased healthcare utilization among patients with chronic CAD.²⁵ The frequency of angina from the patients' perspective is captured with the SAQ AF domain, which has been shown to correlate closely with daily angina diaries.²⁹ Congruent with prior work, angina was categorized as none (SAQ AF score = 100), monthly (SAQ AF score = 61–99), weekly (SAQ AF score = 31–60), and daily (SAQ AF score = 0–30).³⁰ For the purposes of this study, we excluded patients who reported having no angina in the 4 weeks prior to their clinic visit (SAQ AF = 100), as we were interested in investigating the patterns of treatment escalation among patients with chest pain.

Without access to patients' SAQ questionnaires, and immediately after the visit, providers were asked whether the patient had any chest pain in the prior month and, if so, how often the chest pain occurred [daily, weekly, monthly, or less than monthly (less than monthly was considered the same as monthly, for this analysis)]. Under-recognition of angina was defined as the physician reporting a lower frequency of angina frequency than the patient had reported using the SAQ domain (e.g. the patient reported weekly angina and the physician estimated monthly or no symptoms). As patients with correctly recognized and over-recognized angina had similar rates of treatment escalation ($P = 0.23$), they were considered to be a single group.

Treatment escalation

Providers were asked at the end of the visit to specify their CAD treatment plan based on this particular visit. Any of the following were considered treatment escalation: (i) intensification of antianginal medications (increase in dose or addition of a new medication); (ii) ordering a stress test, coronary computed tomography angiography, coronary angiography, or revascularization (PCI or bypass graft surgery); or (iii) hospital admission. Antianginal medications (including doses and frequencies) were recorded at the beginning and end of each clinic visit and included β -blockers, calcium channel blockers, long-acting nitrates, and ranolazine.

Statistical methods

The primary goal of our study was to examine predictors of treatment escalation among patients with CAD and angina and, specifically, whether under-recognition of the patient's angina was associated with treatment escalation by the physician. A secondary aim is to study the variability of treatment escalation among physicians. Demographic and clinical characteristics were compared between patients who did and did not receive treatment escalation using independent t -tests for continuous variables and χ^2 tests for categorical variables. To study predictors of treatment escalation, we constructed a hierarchical multivariable logistic regression model in which both site and physician were included as random effects, to account for the clustering of patients within physicians and within sites. In addition to under-recognition of angina, we included a number of variables that could be expected to be associated with treatment escalation. Variables were selected *a priori* based on literature review and clinical judgement (balanced against concerns of over-fitting) and included the following: age, sex, race (white vs. non-white), self-reported avoiding care due to cost, history of coronary artery bypass graft surgery, SAQ Summary Score, SAQ Angina Stability domain (which is not included in the SAQ Summary Score and reflects recent changes in patients' angina), being on ≥ 2 antianginal medications on arrival to clinic (to account for aggressive antianginal therapy), and provider specialty (interventional vs. non-interventional cardiology). Variability among physicians was explored with a median odds ratio, which estimates the average relative difference in two hypothetically identical patients receiving treatment escalation if seen by two different physicians. For example, median odds ratio of 1 suggests no physician-level variation

in treatment escalation, while an MOR of 2 suggests a two-fold difference in the odds of receiving treatment escalation if two random physicians see an identical patient. All data were complete for the patients in the analytic cohort. All statistical analyses were performed with SAS, version 9.4 (SAS Institute, Inc., Cary, NC, USA).

Results

Patient characteristics

Of 1257 patients with CAD enrolled in APPEAR, evaluated by 155 cardiologists, 411 (32.7%) reported some angina in the prior month and were included in this study. Mean age of patients was 69.0 years, 60.3% were male, and 91.4% were white. Cardiac and non-cardiac co-morbidities were typical of US patients with chronic CAD, with diabetes in 38.2%, hypertension in 80.0%, a history of PCI in 57.2%, a history of myocardial infarction in 38.2%, and a history of coronary artery bypass grafting in 31.6%. Among these patients who all had some angina, most reported monthly symptoms (76.6%), with 18.7% reporting weekly angina and 4.6% reporting daily angina.

Frequency of treatment escalation

Among the 411 patients with CAD who reported angina in the prior month, 106 (25.8%) patients had their treatment escalated at the

end of their visit, of whom 29 (27.3%) were managed with up-titration or addition of antianginal medications, 66 (62.2%) with diagnostic/invasive testing, and 11 (10.4%) with both. The most common pharmacological strategy was prescribing/up-titrating long-acting nitrates followed by prescribing/up-titrating β -blocker therapy (see Supplementary material online, Table S1). The most commonly used diagnostic/invasive evaluation was stress testing, followed by referral for coronary angiography (see Supplementary material online, Table S2). Table 1 shows the demographic and clinical characteristics of the patients stratified by whether the patient received treatment escalation or not. Patients whose treatment was escalated by their physicians were less likely to have a history of myocardial infarction (escalation vs. no escalation: 29.2 vs. 41.3%, $P = 0.027$) and reported lower (worse) disease-specific health status across most of the SAQ domains and the Summary Score.

Association of under-recognition of angina with treatment escalation

Of the 411 patients who reported angina in the prior month, 178 (43.3%) patients were under-recognized by their provider, 88 (21.4%) were correctly recognized, and 145 (35.2%) were believed to have had more angina by physicians than the patients reported. Rates of treatment escalation were similar between patients whose

Table 1 Demographic and clinical characteristics of patients who did and did not have their treatment escalated

	Treatment escalation (n = 106)	No treatment escalation (n = 305)	P-value
Patient factors			
Age (year)	68.3 $\pm$ 11.4	69.2 $\pm$ 11.7	0.465
Male	62.3%	59.7%	0.638
White race	93.1%	90.8%	0.481
Hypertension	84.0%	78.7%	0.241
Heart failure	10.4%	18.4%	0.055
History of myocardial infarction	29.2%	41.3%	0.027
History of coronary stenting	63.2%	55.1%	0.145
History of coronary bypass graft surgery	37.7%	29.5%	0.116
Diabetes mellitus	35.8%	39.0%	0.563
Smoking status			0.419
Never	22.9%	28.1%	
Former	60.0%	58.9%	
Current	17.1%	13.0%	
On ≥ 2 antianginal medications on arrival	43.4%	38.4%	0.361
Self-reported avoidance of care due to cost	7.5%	9.9%	0.472
SAQ Angina Frequency	70.7 $\pm$ 20.4	75.4 $\pm$ 16.7	0.018
SAQ Angina Stability	47.6 $\pm$ 27.1	56.9 $\pm$ 24.8	0.001
SAQ Quality of Life	57.1 $\pm$ 21.5	64.8 $\pm$ 21.1	0.001
SAQ Physical Limitation	59.4 $\pm$ 24.8	59.2 $\pm$ 25.2	0.960
SAQ Treatment Satisfaction	85.3 $\pm$ 14.8	88.8 $\pm$ 14.7	0.038
SAQ Summary Score	62.5 $\pm$ 17.3	66.7 $\pm$ 16.0	0.022
Physician factors			
Male	84.9%	86.6%	0.671
Practice years	20.6 $\pm$ 9.3	20.6 $\pm$ 10.1	0.997
Interventional cardiologist	31.1%	38.0%	0.202

SAQ, Seattle Angina Questionnaire.

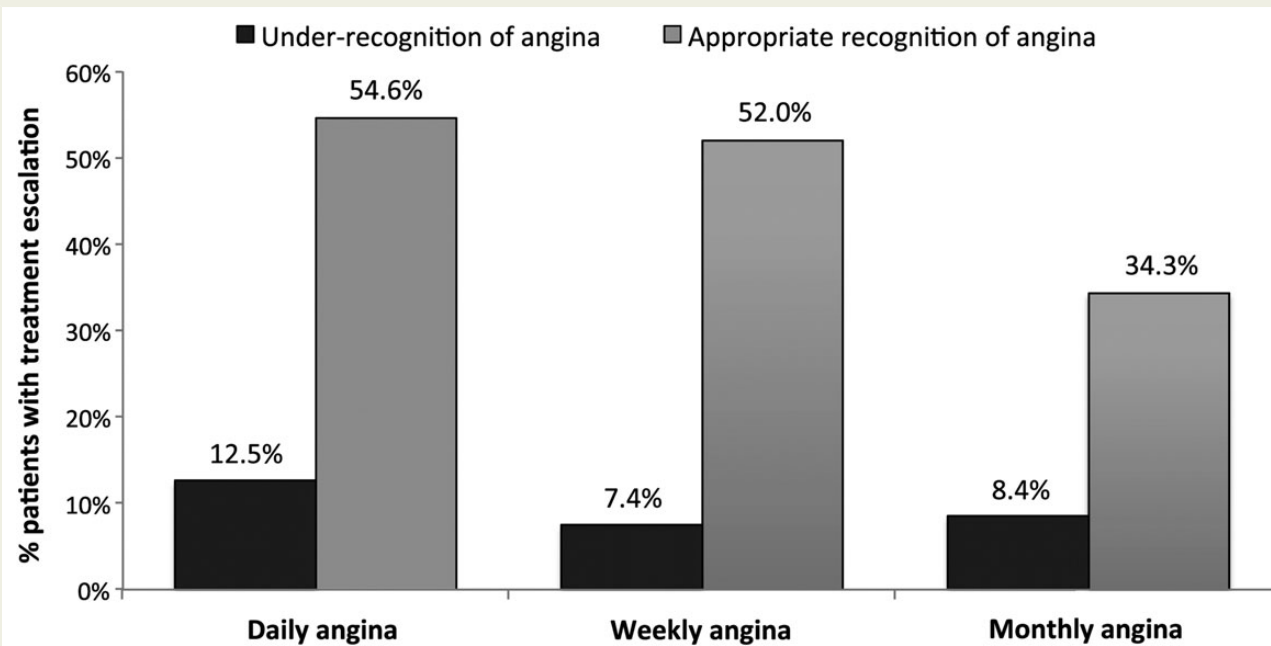


Figure 1 Treatment escalation based on angina frequency and recognition of angina by the physician. *P*-value is <0.001 for all comparisons of under- vs. appropriately recognized angina.

physicians over-estimated their angina (34.1 vs. 42.1%, $P = 0.23$), and thus these two groups were combined into a single group of patients with appropriately recognized angina. Patients with under-recognized angina were less likely to get treatment escalation compared with patients whose angina was appropriately recognized (8.4 vs. 39.1%, $P = <0.001$), and this discrepancy persisted regardless of whether the patient reported daily, weekly, or monthly symptoms (Figure 1; see Supplementary material online, Table S3). For example, among patients with daily angina, only 12.5% of patients with under-recognized daily angina received treatment escalation, when compared with 54.6% of patients whose daily angina was appropriately recognized ($P < 0.001$). In the hierarchical multivariable model that adjusted for demographic and clinical characteristics of the patients (including severity and acceleration of angina), under-recognition continued to be strongly associated with lack of treatment escalation (adjusted OR 0.10, 95% CI 0.04–0.21, $P < 0.001$) (Table 2).

Other factors associated with treatment escalation

Among patients with angina, under-recognition of angina was the only factor significantly associated with treatment escalation. SAQ Angina Stability scores trended towards significance with every 25-point decrease in score associated with a 32% increased odds of treatment escalation (OR 1.32, 95% CI 0.96–1.82, $P = 0.087$), even after adjusting for the burden of angina with the SAQ Summary Score. Notably, age, race, sex, and socio-economic status were not associated with treatment escalation. Whether a patient was on ≥ 2 antianginal medications on arrival was also not associated with treatment escalation (OR 1.17, 95% CI 0.61–2.22, $P = 0.64$). Among the

Table 2 Factors associated with treatment escalation in patients with angina

	OR (95% CI)	P-value
Under-recognition of angina	0.10 (0.04–0.21)	<0.001
Age (per 5 years)	0.91 (0.79–1.04)	0.177
Male	1.04 (0.56–1.95)	0.889
White race	0.39 (0.14–1.07)	0.066
Self-reported avoidance of care due to cost	0.60 (0.20–1.83)	0.365
History of coronary bypass graft surgery	1.16 (0.59–2.27)	0.662
SAQ Angina Stability (per 25 point decrease)	1.32 (0.96–1.82)	0.087
SAQ Summary Score (per 10 point decrease)	0.96 (0.87–1.05)	0.358
On ≥ 2 antianginal medications on arrival	1.17 (0.61–2.22)	0.642
Physician specialty: interventional cardiology	1.09 (0.51–2.34)	0.827

SAQ, Seattle Angina Questionnaire.

subset of patients with daily or weekly angina ($n = 96$), 42 (43.8%) were on < 2 antianginal medications on arrival and thus were eligible for pharmacological or interventional treatment escalation. After seeing the cardiologist, 5 of these patients had their antianginal medications intensified, 8 were referred for diagnostic or invasive testing, and 1 received both interventions, leaving 28 (66.6%) patients with unchanged treatment plans.

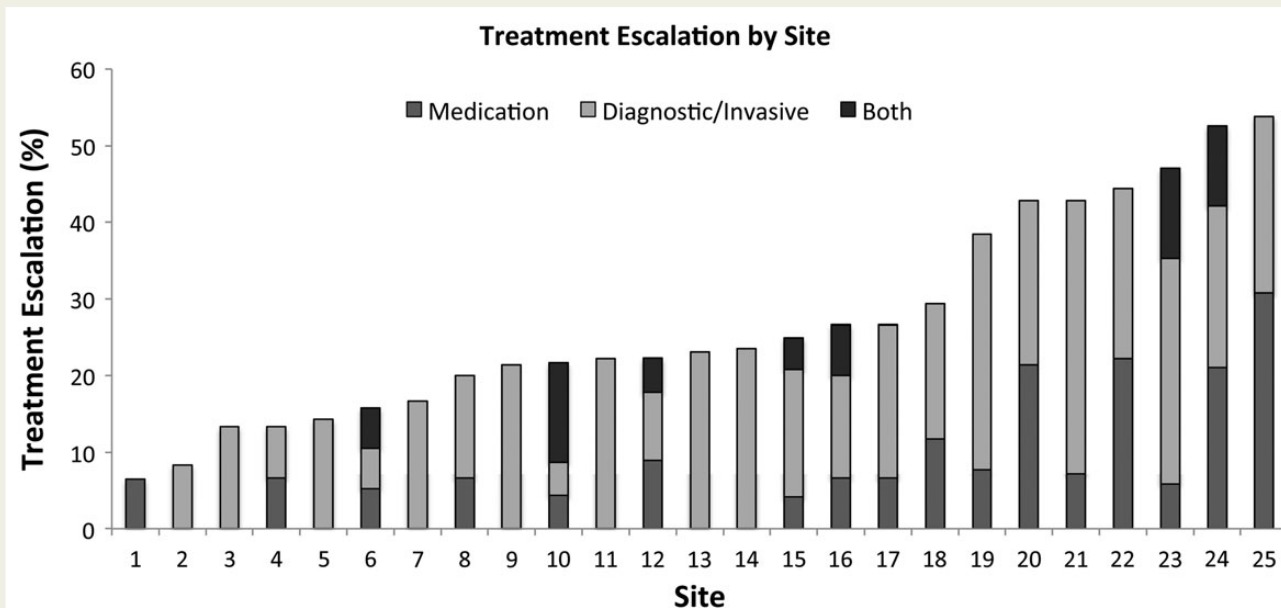


Figure 2 Variability in treatment escalation among different study sites.

In unadjusted analysis, there was substantial variability in the percentage of patients with angina who received treatment escalation and the use of particular treatment strategies (Figure 2). Overall rates of treatment escalation ranged from 6.4 to 53.8%, with a greater reliance on referral for diagnostic or invasive testing for many sites. After adjusting for clinical characteristics and physicians' under-recognition rates, the median odds ratio for variability across physicians was 1.96 ($P = 0.010$), indicating that the odds ratio of treatment escalation varies, on average, two-fold between two randomly selected physicians seeing statistically similar patients.

Discussion

While most investigations on the appropriateness of care in coronary disease have focused on overuse of diagnostic tests and procedures,^{31–35} few have examined underuse of treatment.³⁶ We have previously demonstrated that angina is common in patients with stable CAD and often under-recognized.^{37,38} In this study, we extend our prior work by demonstrating that under-recognition is the factor most strongly associated with not intensifying therapy among patients with angina. Although this association might seem obvious, this is the first multi-center prospective study, to our knowledge, to document the strength of this association, which we believe is striking (i.e. a 10-fold decreased odds of treatment escalation among patients who were under-recognized), and to identify a possible mechanism for the underuse of antianginal strategies. These novel data underscore the importance of testing the implementation of patient-centered outcome measures into routine clinical care to help recognize and quantify patients' angina as a foundation for improving the consistency and quality of care of patients with chronic coronary disease and angina.

The aggressive treatment of angina is a cornerstone of managing patients with coronary disease. Patient-reported chest discomfort is

strongly associated with worse health status, more frequent hospitalizations, and increased healthcare spending.^{25,30} Treatment with antianginal medications or revascularization is known to relieve symptoms and improve quality of life.^{3,5–8} In fact, the magnitude of reduction in angina is the factor most strongly associated with the quality of life benefits from PCI.³⁰ Beyond these traditional treatments, there are additional options for those who continue to have angina despite optimal medical therapy and revascularization.^{14–19} Despite the detrimental effects of frequent angina in patients and the availability of multiple treatment options to improve these symptoms, we found here that whether or not a patient received treatment escalation for their angina did not depend on the patient's clinical history or severity of angina but whether or not the patient's physician appropriately recognized the patient's angina. Patients whose angina was under-recognized were 10 times less likely to be referred for diagnostic or invasive testing or have their antianginal medications intensified compared with patients whose angina was appropriately recognized. We also found that two-thirds of patients with daily or weekly angina who were on <2 antianginal medications (considered to be minimal treatment by the Appropriate Use Criteria³²) were discharged from clinic with unchanged treatment plans. It is important to note that we make no assumption as to what the 'correct' level of treatment escalation should be. However, what we believe is the central message of our study is that there should not be such a discrepancy between rates of treatment escalation simply based on the recognition of angina. In other words, if the 'appropriate' rate of treatment escalation is 50% when the angina is recognized accurately and we observe an adjusted 90% lower rate of treatment escalation when the angina is not accurately recognized, then there are probably missed opportunities for escalating treatment that could represent under-treatment.

Given the frequency with which under-recognition occurs in busy cardiology outpatient practices and the clinical implications

of this under-recognition, these findings support instituting a patient-centered tool for quantifying the burden of patients' angina to facilitate assessment by providers in order to make appropriate treatment and testing recommendations. For example, the SAQ-7²⁸ is a convenient, short, validated, patient-centered tool that could be used to accurately capture patients' angina, when compared with daily diaries.²⁹ The SAQ-7 can be self-administered before the clinic visit, integrated into the electronic medical record, and reviewed by the physician even before entering the exam room, providing a feasible strategy to improve physicians' awareness of patients' angina. Such an approach, analogous to a standardized history of patients' symptoms, could also save time in a busy outpatient clinic and serve as a 'vital sign' in the management of patients' angina. Such a strategy would also be well aligned with the goals of the Centers for Medicare & Medicaid Services in promoting and implementing patient-reported outcome measures to improve the overall quality of care.³⁹ Further studies are needed to understand the effect of these patient-administered tools on both the effectiveness of patient-physician communication and clinical outcomes.

Our study findings should be interpreted with the following limitations. First, although this is a multi-center study from a large number of cardiology outpatient practices around the USA, it is unknown if the findings are generalizable. Second, APPEAR was a cross-sectional study. While we showed that under-recognition of angina was strongly associated with a lack of treatment escalation, we were unable to examine whether these differences in treatment escalation led to differences in health status or other clinical outcomes over time. Other studies are needed to examine the long-term impact of angina under-recognition. Third, the SAQ was completed by patients' right before the clinic visit, which could have made the patients more aware of their symptoms, thereby improving their communication with the physicians. In addition, physicians were aware of the study and that they would be asked to estimate the patient's chest pain and their treatment recommendations. As such, we expect that our findings represent the best-case scenario, and the rates of under-recognition and variability in treatment escalation likely would be higher outside of the confines of a structured study. Alternatively, it is possible that patients believed their physicians would see the SAQ and therefore did not fully report their symptoms. Finally, due to the number of events, we were unable to consider all potential clinical and demographic factors that could be associated with treatment escalation in the model. While we selected the covariates based on clinical judgement and prior literature, it is possible that an important covariate was excluded from the model. Nevertheless, given the strength of the association between under-recognition and treatment escalation, it is unlikely that any additional variable would have materially changed the significance of the identified association.

In conclusion, we found that among patients with CAD and angina in US cardiology outpatient practices, 26% of patients had their anti-anginal medications intensified or were referred for diagnostic or invasive testing. However, this rate varied substantially based on whether or not the physician accurately assessed the frequency of the patient's angina. Further studies are needed to understand the long-term implications of angina under-recognition on health status and other clinical outcomes. Furthermore, studies are needed to test and implement systems that employ health status tools, such

as the SAQ-7, that can objectively assess angina from the patient's perspective and create the ability to readily communicate these data to the physician at the time of treatment.

Supplementary material

Supplementary material is available at *European Heart Journal – Quality of Care and Clinical Outcomes* online.

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References

1. Mozaffarian D, Benjamin EJ, Go AS, Arnett DK, Blaha MJ, Cushman M, de Ferranti S, Despres JP, Fullerton HJ, Howard VJ, Huffman MD, Judd SE, Kissela BM, Lackland DT, Lichtman JH, Lisabeth LD, Liu S, Mackey RH, Matchar DB, McGuire DK, Mohler ER 3rd, Moy CS, Muntner P, Mussolino ME, Nasir K, Neumar RW, Nichol G, Palaniappan L, Pandey DK, Reeves MJ, Rodriguez CJ, Sorlie PD, Stein J, Towfighi A, Turan TN, Virani SS, Willey JZ, Woo D, Yeh RW, Turner MB. American Heart Association Statistics C, Stroke Statistics S. Heart disease and stroke statistics—2015 update: a report from the American Heart Association. *Circulation* 2015;**131**:e29–e322.
2. Brorsson B, Bernstein SJ, Brook RH, Werko L. Quality of life of patients with chronic stable angina before and four years after coronary revascularisation compared with a normal population. *Heart* 2002;**87**:140–145.
3. Weintraub WS, Spertus JA, Kolm P, Maron DJ, Zhang Z, Jurkovic Z, Zhang W, Hartigan PM, Lewis C, Veledar E, Bowen J, Dunbar SB, Deaton C, Kaufman S, O'Rourke RA, Goeree R, Barnett PG, Teo KK, Boden WE, Group CTR, Mancini GB. Effect of PCI on quality of life in patients with stable coronary disease. *N Engl J Med* 2008;**359**:677–687.
4. Fihn SD, Gardin JM, Abrams J, Berra K, Blankenship JC, Dallas AP, Douglas PS, Foody JM, Gerber TC, Hinderliter AL, King SB 3rd, Kligfield PD, Krumholz HM, Kwong RY, Lim MJ, Linderbaum JA, Mack MJ, Munger MA, Prager RL, Sabik JF, Shaw LJ, Sikkema JD, Smith CR Jr, Smith SC Jr, Spertus JA, Williams SV, American College of Cardiology F, American Heart Association Task Force on Practice G, American College of P, American Association for Thoracic S, Preventive Cardiovascular Nurses A, Society for Cardiovascular A, Interventions, Society of Thoracic S. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS Guideline for the diagnosis and management of patients with stable ischemic heart disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, and the American College of Physicians, American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *J Am Coll Cardiol* 2012;**60**:e44–e164.
5. Kerns SJ, Harjai KJ, Stone GW, Grines LL, Boura JA, O'Neill WW, Grines CL. Does beta-blocker therapy improve clinical outcomes of acute myocardial infarction after successful primary angioplasty? *J Am Coll Cardiol* 2004;**43**:1773–1779.
6. Heidenreich PA, McDonald KM, Hastie T, Fadel B, Hagan V, Lee BK, Hlatky MA. Meta-analysis of trials comparing beta-blockers, calcium antagonists, and nitrates for stable angina. *JAMA* 1999;**281**:1927–1936.
7. Abrams J. Clinical practice. Chronic stable angina. *N Engl J Med* 2005;**352**:2524–2533.
8. Arnold SV, Kosiborod M, McGuire DK, Li Y, Yue P, Ben-Yehuda O, Spertus JA. Effects of ranolazine on quality of life among patients with diabetes mellitus and stable angina. *JAMA Intern Med* 2014;**174**:1403–1405.
9. Benzer W, Hofer S, Oldridge NB. Health-related quality of life in patients with coronary artery disease after different treatments for angina in routine clinical practice. *Herz* 2003;**28**:421–428.

10. Wijeyesundera HC, Nallamothu BK, Krumholz HM, Tu JV, Ko DT. Meta-analysis: effects of percutaneous coronary intervention versus medical therapy on angina relief. *Ann Intern Med* 2010;**152**:370–379.
11. Morrison DA, Sethi G, Sacks J, Henderson W, Grover F, Sedlis S, Esposito R, Ramanathan K, Weiman D, Saucedo J, Antakli T, Paramesh V, Pett S, Vernon S, Birjiniuk V, Welt F, Krucoff M, Wolfe W, Lucke JC, Mediratta S, Booth D, Barbieri C, Lewis D, Angina With Extremely Serious Operative Mortality E. Percutaneous coronary intervention versus coronary artery bypass graft surgery for patients with medically refractory myocardial ischemia and risk factors for adverse outcomes with bypass: a multicenter, randomized trial. Investigators of the Department of Veterans Affairs Cooperative Study #385, the Angina With Extremely Serious Operative Mortality Evaluation (AWESOME). *J Am Coll Cardiol* 2001;**38**:143–149.
12. Safley DM, Grantham JA, Hatch J, Jones PG, Spertus JA. Quality of life benefits of percutaneous coronary intervention for chronic occlusions. *Catheter Cardiovasc Interv* 2014;**84**:629–634.
13. Grantham JA, Jones PG, Cannon L, Spertus JA. Quantifying the early health status benefits of successful chronic total occlusion recanalization: results from the Flow-Cardia's Approach to Chronic Total Occlusion Recanalization (FACTOR) trial. *Circ Cardiovasc Qual Outcomes* 2010;**3**:284–290.
14. Verheye S, Jolicœur EM, Behan MW, Pettersson T, Sainsbury P, Hill J, Vrolix M, Agostoni P, Engstrom T, Labinaz M, de Silva R, Schwartz M, Meyten N, Uren NG, Doucet S, Tanguay JF, Lindsay S, Henry TD, White CJ, Edelman ER, Banai S. Efficacy of a device to narrow the coronary sinus in refractory angina. *N Engl J Med* 2015;**372**:519–527.
15. Arora RR, Chou TM, Jain D, Fleishman B, Crawford L, McKiernan T, Nesto RW. The multicenter study of enhanced external counterpulsation (MUST-EECP): effect of EECP on exercise-induced myocardial ischemia and anginal episodes. *J Am Coll Cardiol* 1999;**33**:1833–1840.
16. Soran O, Kennard ED, Kfoury AG, Kelsey SF, Investigators I. Two-year clinical outcomes after enhanced external counterpulsation (EECP) therapy in patients with refractory angina pectoris and left ventricular dysfunction (report from The International EECP Patient Registry). *Am J Cardiol* 2006;**97**:17–20.
17. McNab D, Khan SN, Sharples LD, Ryan JY, Freeman C, Caine N, Tait S, Hardy I, Schofield PM. An open label, single-centre, randomized trial of spinal cord stimulation vs. percutaneous myocardial laser revascularization in patients with refractory angina pectoris: the SPIRiT trial. *Eur Heart J* 2006;**27**:1048–1053.
18. Frazier RH, March RJ, Horvath KA. Transmyocardial revascularization with a carbon dioxide laser in patients with end-stage coronary artery disease. *N Engl J Med* 1999;**341**:1021–1028.
19. Aaberge L, Rootwelt K, Blomhoff S, Saatvedt K, Abdelnoor M, Forfang K. Continued symptomatic improvement three to five years after transmyocardial revascularization with CO₂ laser: a late clinical follow-up of the Norwegian Randomized trial with transmyocardial revascularization. *J Am Coll Cardiol* 2002;**39**:1588–1593.
20. Beltrame JF, Weekes AJ, Morgan C, Tavella R, Spertus JA. The prevalence of weekly angina among patients with chronic stable angina in primary care practices: the Coronary Artery Disease in General Practice (CADENCE) Study. *Arch Intern Med* 2009;**169**:1491–1499.
21. Grodzinsky A, Kosiborod M, Beltrame J, Gosch K, Jones PG, Spertus JA, Alexander KP, Shafiq A. Abstract 18360: predictors of physician under-recognition of angina in outpatients with stable coronary artery disease. *Circ Cardiovasc Qual Outcomes* 2015;**13**:A18360.
22. Burkhardt D, Schmidt S, Schulman SP, Myers J, Resar J, Becker LC, Weiss J, Jones JW. Transmyocardial laser revascularisation compared with continued medical therapy for treatment of refractory angina pectoris: a prospective randomised trial. ATLANTIC Investigators. Angina treatments-lasers and normal therapies in comparison. *Lancet* 1999;**354**:885–890.
23. Brown N, Melville M, Gray D, Young T, Munro J, Skene AM, Hampton JR. Quality of life four years after acute myocardial infarction: short form 36 scores compared with a normal population. *Heart* 1999;**81**:352–358.
24. Spertus JA, Jones P, McDonnell M, Fan V, Fihn SD. Health status predicts long-term outcome in outpatients with coronary disease. *Circulation* 2002;**106**:43–49.
25. Arnold SV, Morrow DA, Lei Y, Cohen DJ, Mahoney EM, Braunwald E, Chan PS. Economic impact of angina after an acute coronary syndrome: insights from the MERLIN-TIMI 36 trial. *Circ Cardiovasc Qual Outcomes* 2009;**2**:344–353.
26. Spertus JA, Winder JA, Dewhurst TA, Deyo RA, Fihn SD. Monitoring the quality of life in patients with coronary artery disease. *Am J Cardiol* 1994;**74**:1240–1244.
27. Spertus JA, Winder JA, Dewhurst TA, Deyo RA, Prodzinski J, McDonnell M, Fihn SD. Development and evaluation of the Seattle Angina Questionnaire: a new functional status measure for coronary artery disease. *J Am Coll Cardiol* 1995;**25**:333–341.
28. Chan PS, Jones PG, Arnold SA, Spertus JA. Development and validation of a short version of the Seattle Angina Questionnaire. *Circ Cardiovasc Qual Outcomes* 2014;**7**:640–647.
29. Arnold SV, Kosiborod M, Li Y, Jones PG, Yue P, Belardinelli L, Spertus JA. Comparison of the Seattle Angina Questionnaire with daily angina diary in the TERISA clinical trial. *Circ Cardiovasc Qual Outcomes* 2014;**7**:844–850.
30. Spertus JA, Salisbury AC, Jones PG, Conaway DG, Thompson RC. Predictors of quality-of-life benefit after percutaneous coronary intervention. *Circulation* 2004;**110**:3789–3794.
31. Bradley SM, Spertus JA, Kennedy KF, Nallamothu BK, Chan PS, Patel MR, Bryson CL, Malenka DJ, Rumsfeld JS. Patient selection for diagnostic coronary angiography and hospital-level percutaneous coronary intervention appropriateness: insights from the National Cardiovascular Data Registry. *JAMA Intern Med* 2014;**174**:1630–1639.
32. Patel MR, Dehmer GJ, Hirshfeld JW, Smith PK, Spertus JA. ACCF/SCAI/STS/AATS/AHA/ASNC/HFSA/SCCT 2012 appropriate use criteria for coronary revascularization focused update: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, Society for Cardiovascular Angiography and Interventions, Society of Thoracic Surgeons, American Association for Thoracic Surgery, American Heart Association, American Society of Nuclear Cardiology, and the Society of Cardiovascular Computed Tomography. *J Am Coll Cardiol* 2012;**59**:857–881.
33. Chan PS, Patel MR, Klein LW, Krone RJ, Dehmer GJ, Kennedy K, Nallamothu BK, Weaver WD, Masoudi FA, Rumsfeld JS, Brindis RG, Spertus JA. Appropriateness of percutaneous coronary intervention. *JAMA* 2011;**306**:53–61.
34. Patel MR, Spertus JA, Brindis RG, Hendel RC, Douglas PS, Peterson ED, Wolk MJ, Allen JM, Raskin IE, American College of Cardiology F. ACCF proposed method for evaluating the appropriateness of cardiovascular imaging. *J Am Coll Cardiol* 2005;**46**:1606–1613.
35. Brindis RG, Douglas PS, Hendel RC, Peterson ED, Wolk MJ, Allen JM, Patel MR, Raskin IE, Hendel RC, Bateman TM, Cerqueira MD, Gibbons RJ, Gillam LD, Gillespie JA, Hendel RC, Iskandrian AE, Jerome SD, Krumholz HM, Messer JV, Spertus JA, Stowers SA, American College of Cardiology Foundation Quality Strategic Directions Committee Appropriateness Criteria Working Group, American Society of Nuclear C, American Heart A. ACCF/ASNC appropriateness criteria for single-photon emission computed tomography myocardial perfusion imaging (SPECT MPI): a report of the American College of Cardiology Foundation Quality Strategic Directions Committee Appropriateness Criteria Working Group and the American Society of Nuclear Cardiology endorsed by the American Heart Association. *J Am Coll Cardiol* 2005;**46**:1587–1605.
36. Ko DT, Guo H, Wijeyesundera HC, Natarajan MK, Nagpal AD, Feindel CM, Kingsbury K, Cohen EA, Tu JV, Cardiac Care Network of Ontario Variations in Revascularization Practice in Ontario Working G. Assessing the association of appropriateness of coronary revascularization and clinical outcomes for patients with stable coronary artery disease. *J Am Coll Cardiol* 2012;**60**:1876–1884.
37. Shafiq A, Kureshi F, Arnold S, Gosch K, Breeding T, Jones P, Spertus J. The prevalence and management of angina among stable CAD patients in outpatient cardiology practices in the United States: insights from the Angina Prevalence and Provider Evaluation of Angina Relief (APPEAR) study. *Circ Cardiovasc Qual Outcomes* 2015;**8**:A111.
38. Shafiq A, Kureshi F, Arnold S, Gosch K, Breeding T, Jones P, Spertus J. Patient and physician discordance in reporting symptoms of angina: insights from the Angina Prevalence and Provider Evaluation of Angina Relief (APPEAR) study. *Circ Cardiovasc Qual Outcomes* 2015;**8**:A7.
39. Prepared by: Health Services Advisory Group I, Title: 2015 National Impact Assessment of the Centers for Medicare & Medicaid Services (CMS) Quality Measures Report, CMS, Baltimore, Maryland, March 2, 2015. <https://www.cms.gov/Medicare/Quality-Initiatives-Patient-Assessment-Instruments/QualityMeasures/QualityMeasurementImpactReports.html> (20 November 2015).